FOR MORPHISMS OF RINGED TOPOI WITH FINITELY GENERATED IDEAL, PUSHFORWARD COMMUTES WITH DERIVED COMPLETION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1723-1736 of sources/stacks-algebraization/pinned-original-source.tex; proof span is lines 1738-1774, the complete author proof (35 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool algebraization wave36 (al02 bib-salvage batch). Spans computed from \label (16.5/16.6).

Editorial note. External \cite keys salvaged with card-local bibliography.tex entries copied verbatim from the snapshot my.bib (precedent S3-morph-blow-up-ample-family); URL/ISSN/EPRINT fields dropped as the preamble has no url package.

Editorial note. statement contains reference environment with real content -> patch-sd-envs.tex applied (sd02 lesson).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; in-chapter lemmas cited by name become stub prereqs; external cites resolved via the card-local bibliography.tex fragment.

Source reference. Original lemma 0A0E: sources/stacks-algebraization/pinned-original-source.tex, source lines 1308–1319. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 099J: sources/stacks-algebraization/pinned-original-source.tex, source lines 1199–1206. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 099F: sources/stacks-algebraization/pinned-original-source.tex, source lines 1515–1520. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A0F: sources/stacks-algebraization/pinned-original-source.tex, source lines 1558–1576. Cited by the retained statement or proof but

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not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A07: sources/stacks-algebraization/pinned-original-source.tex, source lines 1774–1774. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0944: sources/stacks-algebraization/pinned-original-source.tex, source lines 1774–1774. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Generalization of* [1, Lemma 6.5.9 (2)]. *Compare with* [2, Theorem 6.5] *in the setting of quasi-coherent modules and morphisms of (derived) algebraic stacks. Let $f : (Sh(\mathcal{D}), \mathcal{O}') \to (Sh(\mathcal{C}), \mathcal{O})$ be a morphism of ringed topoi. Let $\mathcal{I} \subset \mathcal{O}$ be a finite type sheaf of ideals. Let $\mathcal{I}' \subset \mathcal{O}'$ be the ideal generated by $f^\sharp(f^{-1}\mathcal{I})$. Then Rf_* commutes with derived completion, i.e., $Rf_*(K^\wedge) = (Rf_*K)^\wedge$.*

Proof. By Proposition 099F the derived completion functors exist. By Lemma 099J the object $Rf_*(K^\wedge)$ is derived complete, and hence we obtain a canonical map $(Rf_*K)^\wedge \to Rf_*(K^\wedge)$ by the universal property of derived completion. We may check this map is an isomorphism locally on $\mathcal{C}$. Thus, since derived completion commutes with localization (Remark 0A0F) we may assume that $\mathcal{I}$ is generated by global sections $f_1, \ldots, f_r$. Then $\mathcal{I}'$ is generated by $g_i = f^\sharp(f_i)$. By Lemma 0A0E we have to prove that

$$R\lim \left(Rf_*K \otimes^{\mathbf{L}}_{\mathcal{O}} K(\mathcal{O}, f_1^n, \ldots, f_r^n)\right) = Rf_* \left(R\lim K \otimes^{\mathbf{L}}_{\mathcal{O}'} K(\mathcal{O}', g_1^n, \ldots, g_r^n)\right)$$

Because Rf_* commutes with $R\lim$ (Cohomology on Sites, Lemma 0A07) it suffices to prove that

$$Rf_*K \otimes^{\mathbf{L}}_{\mathcal{O}} K(\mathcal{O}, f_1^n, \ldots, f_r^n) = Rf_* \left(K \otimes^{\mathbf{L}}_{\mathcal{O}'} K(\mathcal{O}', g_1^n, \ldots, g_r^n)\right)$$

This follows from the projection formula (Cohomology on Sites, Lemma 0944) and the fact that $Lf^*K(\mathcal{O}, f_1^n, \ldots, f_r^n) = K(\mathcal{O}', g_1^n, \ldots, g_r^n)$. □

References

[1] Bhatt, Bhargav and Scholze, Peter. *The pro-étale topology for schemes.* 2013. preprint.
[2] Halpern-Leistner, Daniel and Preygel, Anatoly. *Mapping stacks and categorical notions of properness.* 2014.